Optical Comb and Pulse Generation from CW Lightwave

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***Abstract*— Optical comb/pulse sources based on electro-optic modulation techniques are reviewed, where wideband comb and ultrashort pulse train are stably generated from a continuous-wave lightwave. Applications, challenges are discussed focusing on optical and microwave-photonic systems.**

Keywords*— *Optical comb generation, ultra-short pulse generation, electro-optic modulation, Mach-Zehnder modulator, LiNbO3 optical modulator

I. INTRODUCTION

Conventionally, an optical frequency comb or ultra-short pulse train is generated by mode-locked lasers [1]. Recently, electro-optic (EO) modulation with a $LiNbO_3$ waveguide modulator is becoming increasingly attractive for generating an optical frequency comb [2-10] due to improved modulation bandwidth and decreased driving voltage for the modulator [11][12]. This approach enables stable generation of optical comb and pulse with flexible controllability in, for examples, its wavelength and repetition rates (i.e. frequency spacing), and so on [9][10]. In addition, the technologies can offer several attractive applications useful for optical communication, test and measurements [13][14] and microwave photonics [15][16].

In this paper, we review technologies regarding Mach-Zehnder-modulator-based flat comb generator (MZ-FCG), which simply generates optical comb/pulse from continuous-wave light by using single-stage Mach-Zehnder modulator. The optimal operating conditions for flat comb generation are discussed in the half part of this paper. In the latter half of this paper, we review optical short pulse generation using the MZ-FCG. Techniques of ultra-short pulse synthesis from the generated comb are reviewed, including fiber-optic methods based on linear Fourier synthesis and nonlinear pulse compression.

II. MACH-ZENHDER MODULATOR BASED FLAT COMB GENERATOR

Figure 1 shows the concept of the optical frequency comb generation. In the proposed optical frequency comb generator, an input continuous-wave (CW) lightwave is large-amplitude modulated with a conventional MZM, which is dual driven by in-phase sinusoidal signals with different amplitudes.

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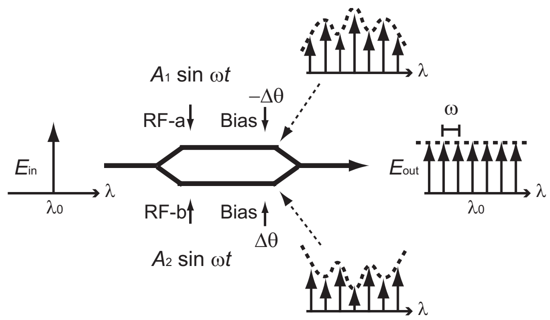

Figure 1: Principle of flat comb generation by using MZ-FCG

Generally, however, the generated comb has a non-flat spectral envelope. We have recently derived the optimal driving condition for flatly generating the frequency comb even using the single-stage MZM setup, which are summarized as the following formulas [6].

Here, in this subsection, principle operation modes for flatly generating optical comb using an MZM are analytically derived.

Suppose that the optical phase shift induced by signals RF-a and RF-b are $\Phi_a(t) = (\overline{A} + \Delta A)\sin(2\pi f_0 t + \Delta\phi_{ab})$, $\Phi_b(t) = (\overline{A} - \Delta A)\sin(2\pi f_0 t - \Delta\phi_{ab})$, respectively, where $\overline{A}$ is the average amplitude of the zero-to-peak phase shift induced by RF-a and Rf-b; $2\Delta A$ is difference between them; f_0 is the modulation frequency; $2\Delta\phi_{ab}$ is the phase difference between RF-a and RF-b.

For large-amplitude driving signals, power conversion efficiency from the input CW light to each harmonic mode can be asymptotically approximated as

$$\begin{aligned}
\eta_k &= \frac{P_k}{P_{in}} \\
&\approx \frac{1}{2\pi\overline{A}}\left| e^{i(\Delta\theta + k\Delta\phi_{ab})}\cos(\alpha + \Delta A) + e^{-i(\Delta\theta + k\Delta\phi_{ab})}\cos(\alpha - \Delta A)\right|^2 \\
&= \frac{1}{2\pi\overline{A}}\left[1 + \cos(2\Delta A)\cos(2\Delta\theta + 2k\Delta\phi_{ab}) + \cos(2\Delta A)\cos\beta\cos(k\pi)\right. \\
&\qquad \left. + \cos(2\Delta\theta + 2k\Delta\phi_{ab})\cos\beta\cos(k\pi)\right] \qquad (1)
\end{aligned}$$

where $\beta \equiv \overline{A} - \frac{\pi}{2}(+\text{Higher} - \text{order term})$. This expression describes behavior of the generated comb well as long as $\overline{A}$ is large enough. Generally, the conversion efficiency is highly

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dependent on the harmonic order of the driving signal, k, which means that the frequency comb generated from the MZM has a non-flat spectrum.

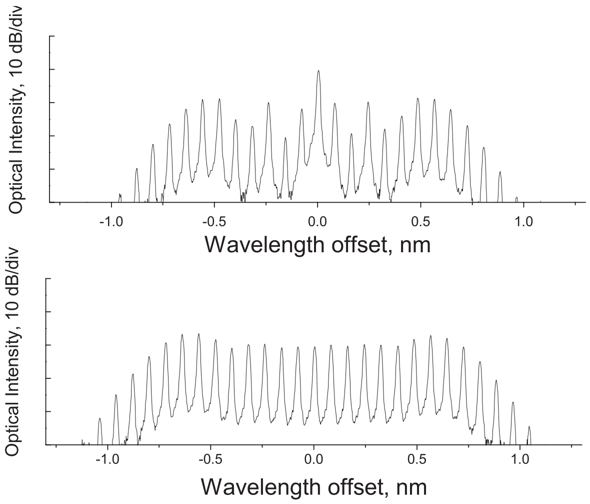

Figure 2: Typical spectra generated from MZ-FCG; (a) single-arm driven, (b) dual-arm drive under the flat-spectrum condition

To make the comb flat in the optical frequency domain, the intensity of each mode should be independent of k. From Eq. 1, the condition is

$$\cos(2\Delta A) + \cos(2\Delta\theta + 2k\Delta\phi_{ab}) = 0 \qquad (2)$$

To keep this equation for any k, the second term should be independent of k. $\Delta\phi_{ab}$ should satisfy

$$\Delta\phi_{ab} = 0 \text{ or } \pm\frac{\pi}{2}. \qquad (3)$$

It should be noted that $\Delta\phi_{ab} = 0$ and $\Delta\phi_{ab} = \frac{\pi}{2}$ correspond to the cases of "in-phase" and "out-of-phase (push-pull)" driven conditions, respectively. In the "in-phase" driven case ($\Delta\phi_{ab} = 0$), the difference of the induced phase difference and bias difference should be related as

$$\Delta A \pm \Delta\theta = n\pi + \frac{\pi}{2}. \qquad (4)$$

to make the spectral envelope flattened.

In the case of $\Delta\phi_{ab} = \frac{\pi}{2}$, the MZM is allowed to be "out-of-phase (push-pull)" driven. From Eq. 3, the flat spectrum condition yields [7]

$$\Delta A = \pm\frac{\pi}{4}, \Delta\theta = \pm\frac{\pi}{4} \qquad (5)$$

From Eq. 4 and Eq. 5, it is found that there are conditions for flat frequency comb generation both for "in-phase" and "out-of-phase" driving cases, and the former condition is more robust since we only need to keep the balance between ΔA and $\Delta\theta$. If we make the efficiency of the generated comb maximum, however, the driving condition for "in-phase" driven case also results in $\Delta A = \pm\frac{\pi}{4}$, $\Delta\theta = \pm\frac{\pi}{4}$.

Figure 2 shows typical optical spectra of the frequency comb generated from the comb generator. Fig. 2(a) indicates the spectrum measured when the modulator was single arm

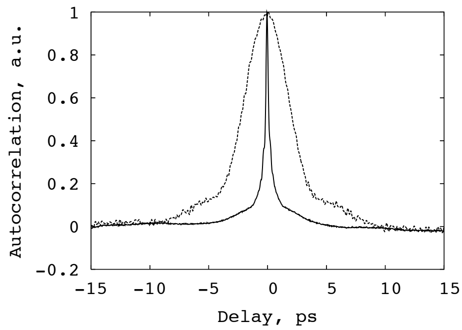

Figure 3: Autocorrelation traces of generated pulse train; Dotted curve: generated by linear pulse synthesis, solid curve: compressed with a dispersion-decreasing fiber

driven only with RF-a; whereas that of Fig. 2(b) was obtained when the modulator was dual-driven with RF-a and RF-b under the driving condition. In the single-driven case, the optical intensity of each mode rippled along the wavelength axis. On the other hand, the modulation spectrum had good flatness when the modulator was dual driven under the "flat-spectrum condition." The frequency spacing and 10-dB spectral width of the generated comb were 10 GHz and 230 GHz, respectively, where the average induced phase shift was 3.2π.

III. SHORT PULSE GENERATION

Generation of picosecond optical pulse train at a high repetition rate has been extensively studied to achieve highly stable and flexible operation, aiming at the use in ultra-high-speed data transmission or in ultra-fast photonic measurement systems. Conventionally, actively/passively mode-locked lasers based on semiconductor or fiber-optic technologies have been typically used to generate such pulse trains. In the technologies, however, the laser cavity should be strictly designed and stabilized to generate stable pulse trains, which reduces flexibility in the operation. Especially, its repetition rate of the generated pulses is almost fixed and its scarce tunability has been provided. In addition, the highly nonlinear properties involved in generating pulses also restrict its operating conditions, which leads to limited output optical power and to uncontrollable chirp characteristics.

In the previous section, ultra-flat frequency comb generation by using only an MZM has been described. In this section, we apply it to generation of ultrafast pulse train. Basically, the strategy to synthesize optical pulse train from the comb source is as follows: (1) Phase differences between frequency components are aligned to be zero to form impulsive pulse train. (2) Profile of temporal waveform is controlled by spectral shaping to the generated comb.

By this approach, pulse trains with a pulse width of picosecond order can be obtained as discussed in this section. These two operations can be achieved in a linear process by simple passive components, as discussed in this section. The first one, phase compensation, is easily achieved by using a commonly used optical dispersive fiber. The second one, spectral shaping, is also achieved with a typical optical bandpass filter. Thin-film filters can be used for this purpose.

In this scheme, the pulse source consists of two sections: one for (a) comb generation and the other for (b) pulse synthesis. Section (a), consisting of a single-stage MZM, has a role to flatly generate a frequency comb. In this section, a continuous-wave (CW) light is EO modulated with the MZM, which is dual-driven by sinusoidal in-phase or out-of-phase signals having different amplitudes. Section (b), on the other hand, is comprised of an optical filter and a fiber, and it spectrally shapes the generated comb into a pulse train having a sync^2-like or a Gaussian-like temporal waveform.

The advantages of this pulse source are 1) the pulses are generated in an optically linear process, so that the optical level of the generated pulse is easily controlled; 2) the pulse source can be started up quickly without the need for complicated control procedures; 3) the repetition rate and the center wavelength of the generated pulse can be flexibly and quickly controlled; 4) the generated pulse train is highly stable due to the simple structure of the pulse generator and to the maturity of the components employed; 5) the pulse generator guarantees ultra-low timing jitter due to the high coherence of the generated comb.

3.1 Theory for pulse synthesis

Here, we discuss synthesis of pulse train from the flat comb generated by MZ-FCG [8].

Phase characteristics of comb

To clarify the phase characteristics of the generated comb, we modify Eq. 1 taking into account of higher-order terms of the output field, yielding

$$E_{\text{out}} = \frac{1}{2}E_{\text{in}}\sum_{k=-\infty}^{\infty}\left[J_k(A_1)e^{j(k\omega t+\theta_1)} + J_k(A_2)e^{j(k\omega t+\theta_2)}\right]$$

$$\approx \frac{E_0}{2}\sqrt{\frac{2}{\pi}}\overline{A}^{-\frac{1}{2}}\sum_{k=-\infty}^{\infty}\left\{\cos\left(\overline{A} - \frac{(2k+1)\pi}{4} + \frac{4k^2-1^2}{8}\overline{A}^{-1} + \Delta A\right)e^{j\Delta\theta}\right.$$

$$\left.\cos\left(\overline{A} - \frac{(2k+1)\pi}{4} + \frac{4k^2-1^2}{8}\overline{A}^{-1} - \Delta A\right)e^{-j\Delta\theta}\right\}e^{j(\overline{\theta}+k\omega t)}, \quad (10)$$

Under the flat spectrum condition for in-phase and out-of-phase modes, respectively, the amplitude and the phase of the frequency modes can be approximated as

$$A_k = \frac{E_0\sin(2\Delta\theta)}{\sqrt{2\pi\overline{A}}}, \Phi_k = \pm\frac{4k^2-1}{8A}, \quad (11)$$

where those series higher than the fourth order series of Φ_k are neglected. It should be noted that the amplitude is independent of the harmonic order of the generated frequency components, k; the optical phases of the modes are related through a parabolic function of k. This equation is valid as long as $|k| << k_{\max} = \pi\overline{A}$ is satisfied.

Linear pulse synthesis by fiber-optic circuits

Next, it is explained how the generated ultra-flat frequency comb is shaped into an ultra-short pulse train in section (b) of the pulse source by using Fourier spectral synthesis. In the case of in-phase operation mode, the story becomes more simple. From Eq. 10**Error! Reference source not found.**, it is found that the optical phase relationship between each mode is in a parabolic function of the mode number. Note that phase compensation with $-\Phi_k$ makes the temporal waveform of the generated comb impulsive. Such a phase compensation can be easily achieved by using a piece of standard optical fiber that gives a parabolic phase shift, i.e., a counter group delay, to the generated comb. The optimal length for the pulse generation is simply obtained as,

$$L = \mp\frac{\partial^2\Phi_k}{\partial k^2}(\beta_2\omega_0^2)^{-1} = \mp(\beta_2\overline{A}\omega_0^2)^{-1}, \quad (12)$$

where β_2 denotes group velocity dispersion in the fiber.

The synthesized pulse train, however, has a rather temporal waveform of the sync^2 function causing a large pedestal around the main pulse, because the generated comb has a rectangular spectrum. In many cases, it is required to shape the temporal waveform of the pulse into Gaussian to suppress the undesired pedestal. If an optical bandpass filter (OBPF) is applied to the generated comb having a cut-off frequency of $f << \frac{1}{2\pi}k_{\max}\omega$, the spectral envelope is shaped into the passband profile of the OBPF; thus, the temporal waveform should be a Fourier transform of the filter passband profile. For instance, if a Gaussian filter is applied to the generated comb together with the appropriate phase compensation of $-\Phi_k$, it is possible to generate Fourier-transform limited Gaussian pulse train with a pulse width of $T = 0.44/f$ and with a repetition of $T_0 = \frac{2\pi}{\omega}$. From this analysis, it is found that the optical pulses can be generated only using linear fiber-optic components. This has numerous practical advantages.

3.2 Picosecond pulse generation

Here describes ultra short pulse generation using the generated frequency comb taking advantage of its good coherence and stability. Experimental setup is shown in Fig. 6. In the pulse generation, the MZM comb generator is followed by the second section comprised of a linear dispersive fiber and an optical bandpass filter. This section linearly synthesizes a pulse train from the comb. We found that the optical phase of each frequency mode of the generated comb is stably fixed and related as a parabolic function along the wavelength axis. This means that a piece of standard optical dispersive fiber easily compensate the optical phase difference to convert the frequency comb to impulsive pulse train. The optical band-pass filter having Gaussian passband helpfully make the temporal waveform Gaussian since the generated comb is spectrally flattened.

The dotted curve in Fig. 2 shows the autocorrelation trace of the generated pulse train. From the autocorrelation trace having half width of 4.1 ps, the pulse width was evaluated as 2.9 ps assuming a Gaussian waveform, where time-bandwidth product was 0.5. The root-mean square timing jitter evaluated from single-sideband phase noise was as low as 130 fs, which almost reached the synthesizer's limit of the driving signal fed to the MZM.

3.3 Femtosecond pulse generation

In order to obtain shorter pulse train in femto-second region, it is effective to adopt the technologies of nonlinear pulse compression together with the picosecond pulse generation. Among the pulse compression techniques, in this experiment, adiabatic soliton compression is adopted, where the generated picosecond pulse train mentioned in the earlier section is introduced into a dispersion decreasing fiber (DDF), where N=1 condition is approximately kept along the fiber; as a result the input pulse train is compressed into shorter one inverse-proportionally to the dispersion. The dotted curve in The solid curve in Fig. 3 shows typically obtained autocorrelation trace of the compressed pulse train. 500fs pulse train was successfully generated [9][10].

III. SUMMARY

In this paper, optical comb/pulse sources based on electro-optic modulation techniques were reviewed, especially focusing on Mazh-Zehnder modulator Based Flat Comb Generator (MZ-FCG).

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